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Team ID	N/A
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
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FinRelief AI: AI-Powered Debt Relief & Financial Recovery Platform

Project Description:

FinRelief AI harnesses a rule-based Financial Health Score engine together with Generative AI to help users understand and recover their financial health. The platform includes a Debt & Expense Tracker, which consolidates all of a user's liabilities and spending in one place; a Financial Health Score module, which computes a 0-100 score from debt-to-income, expense, and savings ratios; and an AI Debt Analysis & Assistant, which delivers personalized, data-grounded financial guidance and an interactive chat experience.

Utilizing Google's Gemini API, FinRelief AI processes a user's real financial data to deliver fast, high-quality, personalized advice. Built with FastAPI and React (Vite), and deployed publicly via Render, the platform ensures a seamless and user-friendly experience. With secure JWT-based authentication, bcrypt password hashing, and responsible input handling, FinRelief AI empowers everyday users to move from financial anxiety to a clear, actionable path toward becoming debt-free.

Scenarios

Scenario 1: A user registers, sets up a financial profile with income and expenses, and adds their existing debts. The dashboard immediately shows total debt, credit utilization, and a computed Financial Health Score, all in seconds.

Scenario 2: A user requests AI Debt Analysis. The system reviews their real financial profile via the Gemini API and returns personalized advice on prioritizing repayments and reducing discretionary expenses.

Scenario 3: A user opens the Repayment Planner, selects the Avalanche method, and receives a month-wise plan with an estimated debt-free date, then downloads a PDF summary report to track their progress offline.

Technical Architecture:

Description: FinRelief AI utilizes a modular three-tier architecture to deliver rapid, AI-driven financial insights. The frontend provides a responsive, chart-driven user interface built with React (Vite) and Recharts, while the FastAPI-based backend orchestrates data validation, financial score computation, and prompt construction. It interfaces with the Google Gemini API using the gemini-1.5-flash model to generate tailored financial guidance, with data persisted via SQLAlchemy ORM and both services deployed publicly on Render.

(Note: The database is relational (SQLite, configurable for PostgreSQL via SQLAlchemy); core tables include Users, Financial Profiles, Debts, Expenses, AI Reports, Financial Scores, and Repayment Plans.)

Pre-requisites:

- FastAPI Framework Knowledge: [FastAPI Documentation](#)
- Google Gemini API Familiarity: [Google AI Studio Documentation](#)
- React (Vite), HTML, CSS Skills: [Vite / React Documentation](#)
- Python Programming Proficiency: [Python Documentation](#)
- SQLAlchemy ORM: [SQLAlchemy Documentation](#)
- Version Control with Git: [Git Documentation](#)
- Development Environment Setup: [Visual Studio Code Documentation](#)

Project Workflow:

Milestone 1: Model Selection and Architecture

- Activity 1.1: Generate a Google Gemini API key and configure it securely in the project environment.
- Activity 1.2: Research and select the appropriate Gemini model for personalized financial analysis and chat.
- Activity 1.3: Define the architecture of the application, detailing interactions between the frontend, backend, and AI integration.
- Activity 1.4: Set up the development environment, installing necessary libraries and dependencies for FastAPI and the Gemini API.

Milestone 2: Core Backend Development

- Activity 2.1: Develop the core data models (Users, Financial Profiles, Debts, Expenses, AI Reports, Financial Scores, Repayment Plans).
- Activity 2.2: Implement the FastAPI routers for auth, profile, debts, expenses, dashboard, ai, and reports, ensuring smooth request handling and JWT-protected access.
- Activity 2.3: Build the rule-based Financial Health Score engine and integrate the Gemini client for AI-driven insights.

Milestone 3: App Core Logic Development

- Activity 3.1: Write the main FastAPI application logic in main.py, establishing routes for each feature and integrating AI responses.
- Activity 3.2: Implement input validation and error handling, including an offline demo mode that returns placeholder AI responses when no GEMINI_API_KEY is configured.

Milestone 4: Frontend Development

- Activity 4.1: Design and develop the user interface using React (Vite), ensuring a responsive and intuitive layout for Login/Register, Dashboard, Debt Management, and Expense Tracker pages.
- Activity 4.2: Integrate Recharts to visualize spending breakdowns and debt distribution, and connect the frontend to the backend via an Axios API client.

Milestone 5: AI & Repayment Features

- Activity 5.1: Implement the AI Debt Analysis and AI Financial Assistant features using the Gemini API.
- Activity 5.2: Build the Snowball/Avalanche Repayment Planner with month-wise projections and an estimated debt-free date.
- Activity 5.3: Implement PDF financial report generation using ReportLab.

Milestone 6: Testing and Deployment

- Activity 6.1: Perform functional and performance testing on core APIs using Postman.
- Activity 6.2: Deploy the backend and frontend separately on Render.
- Activity 6.3: Validate all features end-to-end on the live deployment.

Activity 1.1: Configure Google Gemini API Key

- Step 1 – Visit Google AI Studio: Go to <https://aistudio.google.com/> and sign in with a Google account.
- Step 2 – Navigate to API Keys: Click on "Get API Key" from the Google AI Studio menu.
- Step 3 – Create a New API Key: Click the "Create API Key" button and select or create a Google Cloud project.
- Step 4 – Copy the Key: Copy the generated key immediately and store it safely – it will not be shown again.
- Step 5 – Add the Key to Your Project: Create a .env file in the backend root directory and add:
GEMINI_API_KEY=your_api_key_here
- Step 6 – Never Commit Your API Key: Add .env to your .gitignore file to prevent accidentally pushing the key to a public repository.

Exploring the Website's Web Pages:

Dashboard Page: The central hub of FinRelief AI, displaying total debt, total monthly expenses, credit utilization, and the computed Financial Health Score alongside spending and debt-distribution charts.

Debt & Expense Management: Allows users to add, update, and delete debts and expenses, each categorized for clearer tracking.

AI Financial Assistant: An interactive chat interface where users can ask financial questions and receive personalized, data-grounded responses powered by the Gemini API.

Repayment Planner: Lets users choose between the Snowball and Avalanche repayment strategies and view a month-wise plan with an estimated debt-free date.

Conclusion:

FinRelief AI demonstrates how a rule-based scoring engine can be combined with Generative AI and a modular full-stack architecture to help users understand and improve their financial health. The project, which included model selection, backend/frontend development, and deployment via Render, highlights how targeted AI and a structured framework can meaningfully reduce financial stress for everyday users. Looking ahead, FinRelief AI offers significant opportunities for expansion, such as Credit Score Prediction, OCR-based bank statement analysis, a voice-enabled AI assistant, and multilingual support.